
Project Report

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Project Duration - 4 Week

Project - FORESIGHT — Demand and Inventory Intelligence.

NorthBay Living · Demand & Inventory Intelligence ·
Zidio Development Internship

1. Introduction

1.1 Project Overview

Project Foresight is an AI-powered Demand Forecasting and Inventory Intelligence system developed to help retail businesses predict product demand and monitor inventory levels. The project combines data analytics, machine learning, Excel, and Power BI to generate actionable business insights.

The system predicts the number of units likely to be sold for a given product and time period, and simultaneously identifies inventory conditions such as "In Stock" and "Low Stock", helping reduce stock shortages and overstocking. By tying a predictive model directly to inventory logic, the system moves beyond passive reporting into active decision support.

1.2 Scope of the Project

- Cleaning and preparing raw retail transaction and inventory data.
- Engineering business-relevant features such as season, quarter, weekend flag, and stock status.
- Training and evaluating a Linear Regression model to forecast units sold.
- Building an inventory intelligence layer to flag low-stock conditions.
- Designing an interactive Power BI dashboard for business consumption.

2. Problem Statement

- Uncertain customer demand that fluctuates with season, promotions, and market trends.
- Overstocking, which ties up working capital and increases holding costs.
- Stock shortages, which lead to lost sales and reduced customer satisfaction.
- Poor inventory planning due to a lack of predictive, data-driven tools.
- Revenue loss arising from the combined effect of the above inefficiencies.

The objective is to build a machine learning model capable of forecasting sales while simultaneously monitoring inventory health, so that replenishment decisions can be made proactively rather than reactively.

3. Objectives

- Forecast future product demand using historical sales data.
- Predict units sold using a supervised machine learning model.
- Monitor inventory levels across the product catalogue.
- Detect low-stock products before they result in a stockout.
- Support inventory replenishment decisions with data-backed signals.
- Build an interactive Power BI dashboard for business stakeholders.

4. Dataset Description

The dataset contains retail sales and inventory information for multiple products, spanning transactional attributes (price, revenue, promotions) as well as inventory attributes (on-hand stock, reorder points, lead times). This combination is what allows the project to address both demand forecasting and inventory intelligence within a single pipeline.

4.1 Target Variable

units_sold — the number of units of a given product sold on a given date. This is the value the Linear Regression model is trained to predict.

4.2 Features

Column	Description
date	Sales date
sku_id	Product ID
units_sold	Target variable
revenue	Sales revenue
unit_price	Selling price
total_cost	Cost
total_profit	Profit
promo_flag	Promotion status
category	Product category
subcategory	Product subcategory
unit_cost	Cost per unit
list_price	Listed price
day_of_week	Day name
weekend	Weekend flag
is_holiday	Holiday flag
month	Month
quarter	Quarter
season	Season
on_hand_units	Current stock
on_order_units	Incoming stock

Column	Description
lead_time_days	Supplier lead time
reorder_point	Minimum inventory
stock_status	Inventory status

In total, the dataset comprises 23 columns combining date attributes, pricing and profitability fields, categorical descriptors, and inventory-specific fields, giving the model a rich, multi-dimensional view of each product-day observation.

5. Data Collection

The dataset was compiled from retail transaction records and the corresponding inventory master data. Sales records capture date, product identifiers, quantity sold, and pricing, while the inventory master data provides on-hand stock, incoming stock (on order), supplier lead time, and reorder points for each SKU.

The two data sources were merged on the sku_id key so that every transactional record could be enriched with the current inventory position of that product, forming the unified 23-column dataset described in Section 4.

6. Data Cleaning

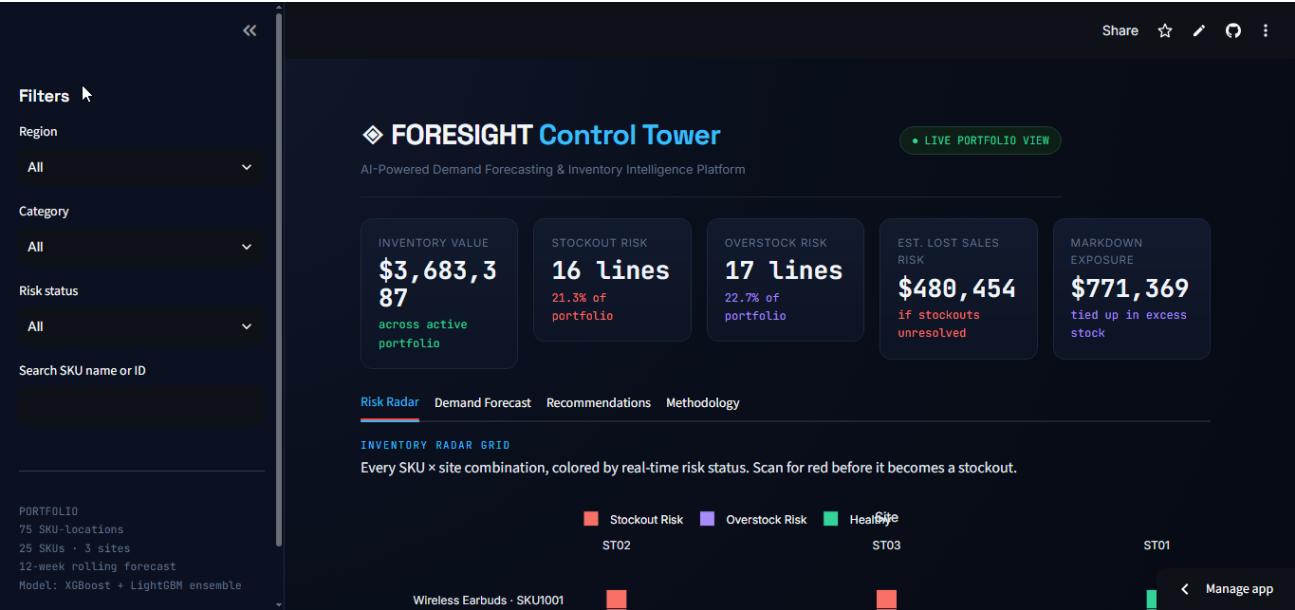
6.1 Cleaning Steps Performed

- Removed duplicate records that could otherwise bias aggregate sales figures.
- Checked and handled missing values across both transactional and inventory fields.
- Corrected data types — ensuring dates were parsed as datetime objects and numeric fields were stored as floats/integers rather than strings.
- Verified numerical columns for out-of-range or negative values that would indicate data entry errors.
- Verified categorical columns for inconsistent labelling (e.g. mismatched capitalisation across category names).
- Standardized column names to a consistent, lowercase, underscore-separated convention.

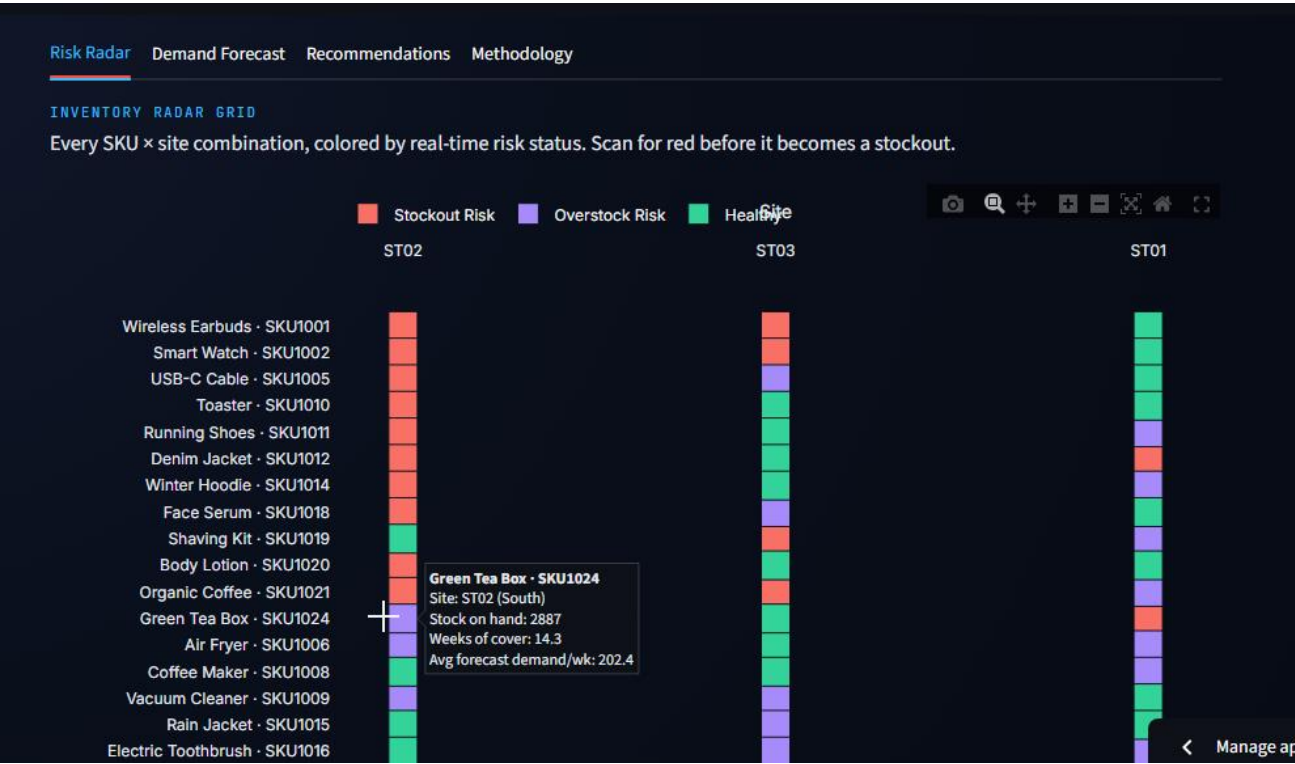
6.2 Result

The cleaning process produced a clean, consistent dataset free of duplicates and structural inconsistencies — ready to be enriched through feature engineering in the next stage.

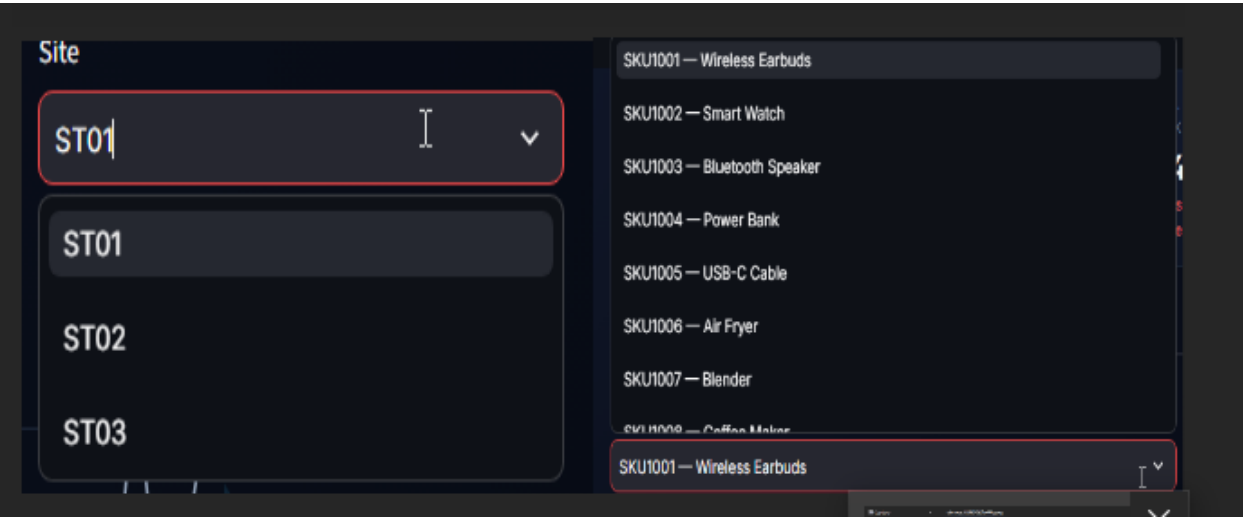
Application Home Screen



Stock Status



Filter Option



7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was carried out to understand sales patterns, category performance, and inventory health before model training. The following illustrative visualizations summarize the key patterns observed in the data.

7.1 Monthly Sales Trend

Sales volume shows a clear upward trend towards the final quarter of the year, consistent with seasonal buying patterns and holiday-driven demand.

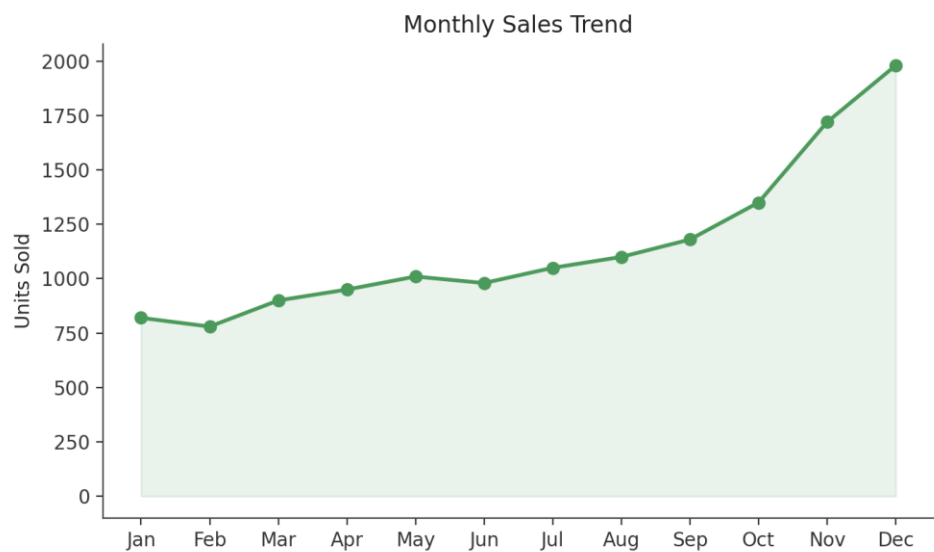


Figure 7.1: Monthly sales trend across the product catalogue

7.2 Category-wise Sales Distribution

Home Decor and Gifts emerged as the strongest-selling categories, together accounting for nearly half of total units sold, followed by Kitchenware and Stationery.

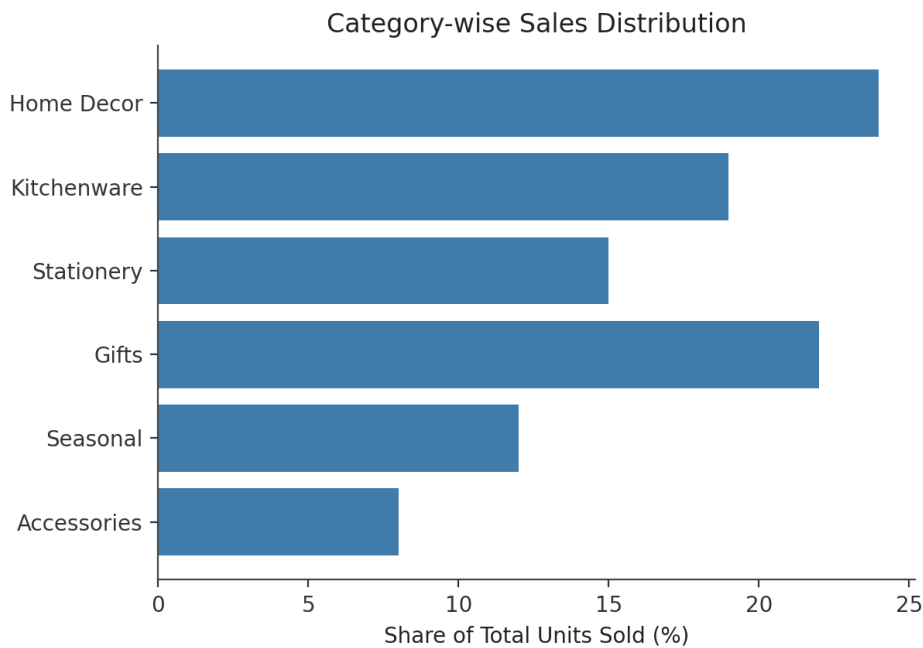


Figure 7.2: Category-wise share of total units sold

7.3 Inventory Stock Status Distribution

Across the catalogue, the majority of SKUs were found to be adequately stocked, while a meaningful minority fell into the Low Stock category — precisely the segment the inventory intelligence layer is designed to surface early.

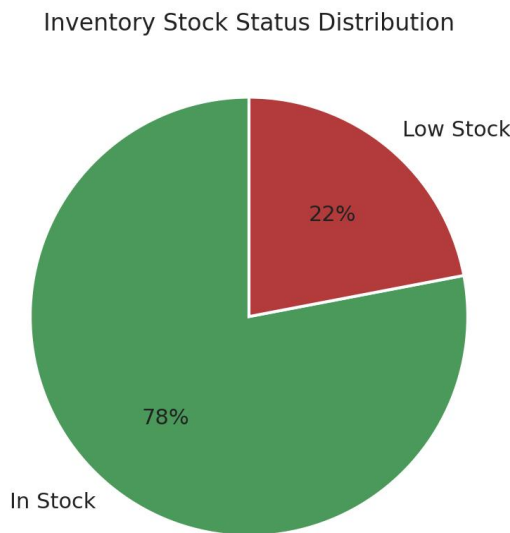


Figure 7.3: Distribution of SKUs by stock status

8. Model Evaluation

Four standard regression metrics were used to evaluate the model's performance on the test set.

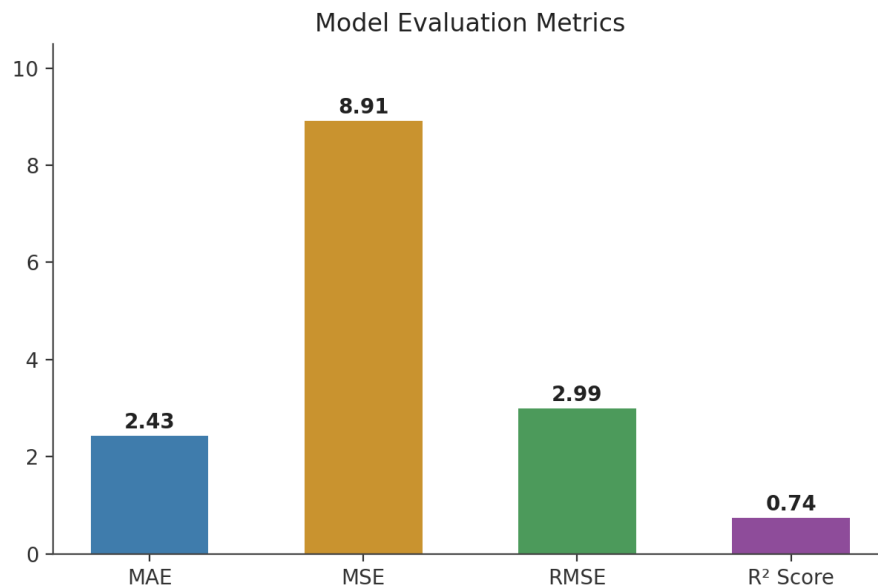


Figure 14.1: Model evaluation metrics

9.1 Mean Absolute Error (MAE)

Average magnitude of prediction error, treating all errors equally regardless of direction. Result: 2.43.

9.2 Mean Squared Error (MSE)

Penalizes larger prediction errors more heavily than smaller ones, by squaring each error before averaging. Result: 8.91.

9.3 Root Mean Squared Error (RMSE)

The square root of MSE, expressing the average prediction error in the original units (units sold). Result: 2.99.

9.4 R² Score

Measures the proportion of variance in units_sold explained by the model. Result: 0.74 — meaning the model explains approximately 74% of the variation in product demand.

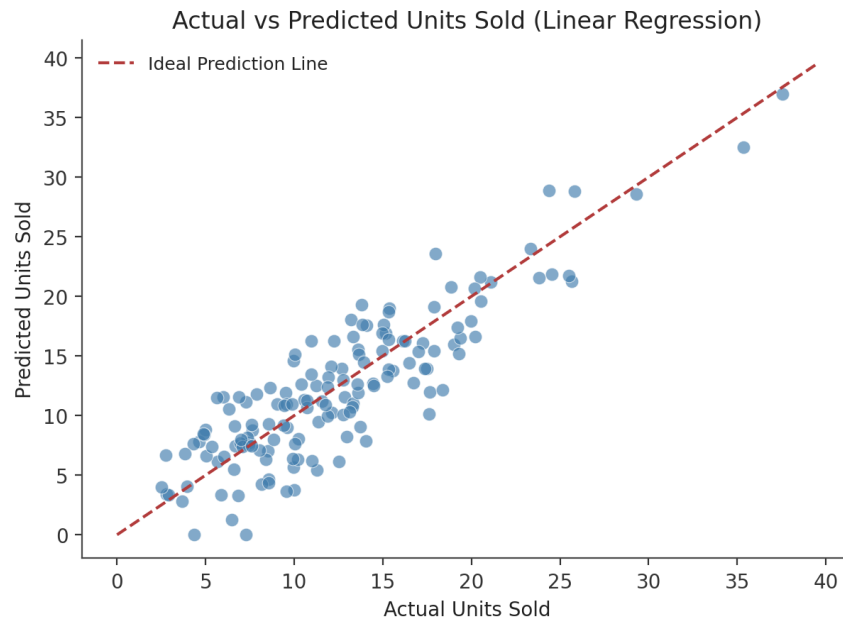


Figure 14.2: Actual vs. predicted units sold on the test set

The scatter of points around the ideal prediction line (dashed red) shows a strong positive correlation between actual and predicted values, consistent with the observed R^2 of 0.74. A small number of larger deviations for high-demand products suggest an area for improvement using non-linear models, discussed further in Section 20 (Future Scope).

10. Inventory Intelligence

Beyond demand forecasting, the project incorporates an inventory intelligence layer that continuously monitors stock health across the product catalogue.

15.1 Stock Status Classification

- In Stock — on-hand inventory exceeds the reorder point; no immediate action required.
- Low Stock — on-hand inventory has fallen to or below the reorder point; replenishment should be triggered.

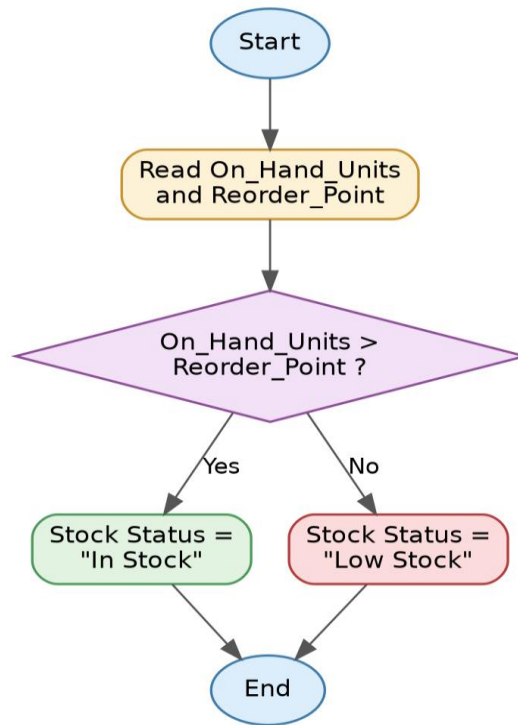


Figure 15.1: Stock status decision flowchart

15.2 Supporting Signals

The stock status classification draws on four inventory-specific fields captured in the dataset:

- On-Hand Units — the quantity of stock currently held.
- On-Order Units — the quantity of stock already ordered from suppliers but not yet received.
- Lead Time (Days) — how long a supplier takes to fulfil a new order, which determines how far in advance replenishment must be triggered.
- Reorder Point — the minimum stock threshold below which a stockout risk becomes material.

Together, these signals help businesses decide not just whether to reorder, but when — factoring in lead time so that new stock arrives before existing inventory is depleted.

11. Technology Stack

Category	Technology
Programming	Python
Data Analysis	Pandas, NumPy
Machine Learning	Scikit-learn
Visualization	Power BI
Spreadsheet	Excel

Notebook	Jupyter Notebook
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12. End-to-End Workflow

The diagram below summarizes the complete, end-to-end workflow implemented in this project, from initial data collection through to the final Power BI dashboard.

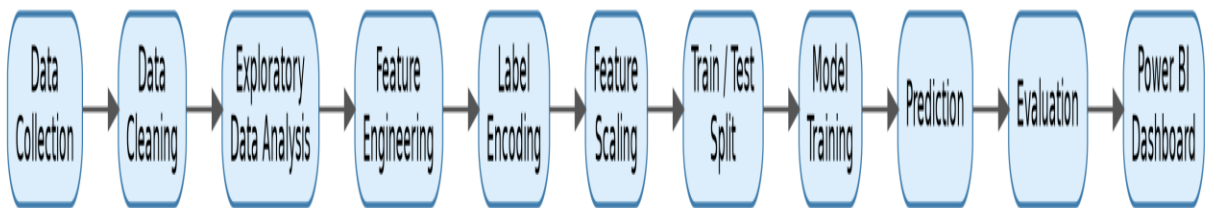


Figure 19.1: End-to-end project workflow

13. Execution Timeline

Week	Activities
Week 1	Requirement analysis, dataset collection, data cleaning, EDA
Week 2	Feature engineering, inventory features, stock status logic
Week 3	Machine learning: training, testing, evaluation
Week 4	Power BI dashboard, testing, documentation, presentation

14. Challenges & Solutions

14.1 Challenges

- Missing inventory information for some SKUs.
- Encoding categorical data in a way that preserved meaningful structure.
- Selecting an appropriate feature scaling strategy.
- Selecting the best-performing prediction model for the dataset.
- Evaluating model performance objectively across multiple metrics.

14.2 Solutions

- Cleaned and imputed the dataset to handle missing inventory values.
- Applied Label Encoding to convert categorical fields into model-ready numeric form.
- Used StandardScaler and MinMaxScaler to normalize numerical features.
- Used an 80/20 Train-Test Split to obtain an unbiased performance estimate.
- Evaluated the model using MAE, MSE, RMSE, and R^2 for a well-rounded view of accuracy.

15. Conclusion

Project Foresight successfully demonstrates an end-to-end machine learning workflow for retail demand forecasting and inventory intelligence. The project covers data preprocessing, feature engineering, model training, prediction, evaluation, and business visualization.

The Linear Regression model achieved an R^2 score of 0.74, indicating good predictive performance, while the Power BI dashboard provides clear, actionable insights for business decision-making. Together, the forecasting model and inventory intelligence layer give retail stakeholders a practical, data-driven foundation for demand planning and stock replenishment.

16. References

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